

Summary

Ref	Year	Focus Area	Methods Used	Key Contribution	Limitation
[1]	2022	Skin lesion classification	CNN, Transfer learning	High accuracy on HAM10000	No segmentation
[2]	2025	Skin cancer diagnosis	Deep CNN	Improved melanoma detection	High computation cost
[3]	2025	Medical image analysis	Hybrid DL	Robust feature extraction	Not real-time
[4]	2024	Skin lesion segmentation	U-Net variants	Accurate boundary detection	CNN-only approach
[5]	2025	Skin disease detection	Deep learning	Improved generalization	Dataset-dependent
[6]	2025	Clinical decision support	ML + imaging	Diagnostic assistance	No real-time use
[7]	2021	Skin lesion analysis	CNN ensemble	Reduced misclassification	Large model size
[8]	2022	Melanoma detection	CNN + preprocessing	Noise-robust detection	No segmentation
[9]	2024	Skin cancer screening	Transformer-based DL	Global feature modeling	High latency
[10]	2025	Medical image segmentation	Attention U-Net	Precise segmentation	Complex architecture
[11]	2025	Skin lesion detection	DL pipeline	Improved sensitivity	Not lightweight
[12]	2024	Medical image classification	ML + DL	Comparative performance study	No hybrid models
[13]	2022	Lesion segmentation	CNN	Strong pixel accuracy	Poor generalization
[14]	2023	Skin image analysis	DL + explainability	Better interpretability	Lower speed
[15]	2022	Skin lesion diagnosis	CNN + augmentation	Improved robustness	CNN limitations
[16]	2023	Disease segmentation	Transformer models	Long-range dependency capture	High computation
[17]	2022	Skin lesion classification	Lightweight CNN	Faster inference	Lower accuracy
[18]	2022	Medical image AI	DL framework	Clinical relevance	Not task-specific
[19]	2023	Image segmentation	Hybrid DL	Improved boundary precision	Training complexity
[20]	2022	Skin cancer detection	CNN	High classification accuracy	No efficiency analysis
[21]	2022	Medical imaging	CNN-based DL	Feature learning	No transformer usage

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Ref	Year	Focus Area	Methods Used	Key Contribution	Limitation
[22]	2021	Skin lesion analysis	ML + DL	Early diagnosis	Limited datasets
[23]	2020	Medical image processing	DL	Automation support	Low scalability
[24]	2020	Image segmentation	CNN	Accurate masks	Not real-time
[25]	2022	Healthcare AI	ML survey	Research overview	No implementation
[26]	2020	Disease detection	ML models	Diagnostic assistance	Outdated models
[27]	2021	Skin lesion segmentation	CNN	Pixel-level precision	High inference time
[28]	2021	Medical AI systems	DL framework	End-to-end design	No efficiency focus
[29]	2018	Image classification	CNN	Foundational work	Old architecture
[30]	2024	Medical image analysis	Advanced DL	Improved accuracy	Not real-time

Our Main Goal

Detect AND segment skin cancer in real time using an efficient ViT-based framework.

Previous studies on skin cancer analysis mainly focus on either classification or segmentation using convolutional neural networks, with limited attention to real-time performance and integrated diagnostic frameworks. Although recent works have introduced Vision Transformers, their use is often restricted to classification tasks without efficient segmentation or deployment considerations. In contrast, this research proposes an efficient Vision Transformer-based framework combined with a U-Net architecture to perform real-time skin cancer detection and segmentation simultaneously. The proposed approach aims to improve clinical usefulness by providing both diagnostic decisions and precise lesion localization while maintaining fast inference speed suitable for real-world applications.

Summary

Limitations :

Despite significant progress in AI-based disease detection and medical image analysis, existing research still faces several important limitations.

First, most studies rely on limited and highly curated datasets, often collected under controlled conditions. Public datasets such as HAM10000 provide valuable benchmarks, but they do not fully represent real-world clinical variability such as differences in skin tone, lighting conditions, imaging devices, and lesion types. This limits the generalization ability of proposed models in real clinical environments.

Second, many approaches focus on either classification or segmentation, but not both simultaneously. While classification models identify whether a lesion is malignant or benign, they often fail to localize precise lesion boundaries. Conversely, segmentation-only models do not provide diagnostic decisions. The lack of unified frameworks that perform detection and segmentation together reduces clinical usefulness.

Third, computational complexity remains a major challenge, especially for real-time applications. Several deep learning and transformer-based models achieve high accuracy but require heavy computational resources, making them unsuitable for deployment on edge devices, mobile platforms, or resource-constrained healthcare settings.

Fourth, limited attention is given to real-time performance and latency analysis. Many studies report accuracy metrics such as precision, recall, and Dice score but do not evaluate inference speed, memory consumption, or real-time feasibility, which are critical for practical medical decision support systems.

Fifth, lack of explainability and clinical interpretability is another key limitation. Most deep learning models act as black boxes, offering limited insight into why a particular prediction is made. This reduces trust among clinicians and restricts adoption in medical practice.

Finally, insufficient external validation and clinical trials weaken the reliability of existing systems. Most models are evaluated using cross-validation on a single dataset, with very few studies testing performance across multiple institutions or real hospital data.

Summary

Future Work

While the proposed Vision Transformer–based framework demonstrates promising performance for real-time skin cancer detection and segmentation, several directions can be explored to further enhance its effectiveness and clinical applicability.

First, future research can focus on **expanding and diversifying datasets** by incorporating multi-institutional and multi-ethnic dermoscopic images. Including a wider range of skin tones, rare lesion types, and early-stage cancers will improve model generalization and reduce dataset bias.

Second, **multi-modal learning** can be investigated by combining dermoscopic images with clinical metadata such as patient age, gender, lesion location, and medical history. Integrating visual and non-visual information may significantly enhance diagnostic accuracy and robustness.

Third, to improve real-time deployment, future work should explore **model optimization techniques** such as pruning, quantization, and knowledge distillation. These methods can reduce computational complexity and memory usage, enabling efficient deployment on mobile devices, edge hardware, and low-resource clinical settings.

Fourth, extending the framework to support **cross-modality generalization**, including clinical photographs and smartphone images, would increase its practical usability beyond dermoscopic imaging. Domain adaptation and transfer learning strategies can be applied to address this challenge.

Fifth, future studies should emphasize **model explainability and interpretability** by integrating attention visualization, saliency maps, or explainable AI (XAI) techniques. Providing transparent decision-making mechanisms will increase clinician trust and support adoption in real medical environments.

Finally, **large-scale clinical validation and real-world testing** are essential future steps. Conducting prospective studies, collaborating with hospitals, and evaluating system performance in real clinical workflows will help bridge the gap between research and practical healthcare deployment.

Summary

TOPIC:

Efficient Deep Learning Framework for Real-Time Skin Cancer Detection and Segmentation with Vision Transformers

Model Used in This Research

Hybrid Vision Transformer + CNN-based Segmentation Model

Vision Transformer (ViT) + U-Net–style Decoder

This research uses a hybrid deep learning model that combines a Vision Transformer (ViT) as the encoder with a U-Net–based decoder for simultaneous skin cancer detection and segmentation.

This research uses a hybrid deep learning model combining a Vision Transformer–based encoder with a U-Net–style decoder. The Vision Transformer is employed to capture global contextual features and complex lesion patterns, while the U-Net decoder performs precise pixel-level segmentation of skin cancer regions. This combination enables simultaneous detection and segmentation, improves accuracy, and supports real-time performance. The model is well-suited for dermoscopic image analysis and addresses limitations of traditional CNN-based approaches.

Summary

Paper link :

- 1)<https://www.mdpi.com/2076-3417/12/12/5990>
- 2)<https://www.nature.com/articles/s41598-025-89230-7>
- 3)<https://www.sciencedirect.com/science/article/pii/S2590005625001420>
- 4)https://link.springer.com/chapter/10.1007/978-3-031-46341-9_6
- 5)<https://www.nature.com/articles/s41598-025-88753-3>
- 6)<https://www.sciencedirect.com/science/article/abs/pii/S1746809425001387>
- 7)<https://ieeexplore.ieee.org/abstract/document/9393198>
- 8)<https://www.mdpi.com/2075-4418/12/9/2115>
- 9)<https://ieeexplore.ieee.org/abstract/document/10993996>
- 10)<https://www.sciencedirect.com/science/article/pii/S0010482525010753>
- 11)<https://www.nature.com/articles/s41598-025-31816-2>
- 12)<https://onlinelibrary.wiley.com/doi/full/10.1155/2024/3022192>
- 13)<https://www.sciencedirect.com/science/article/pii/S0010482522006746>
- 14)<https://link.springer.com/article/10.1007/s11517-023-02969-x>
- 15)<https://www.mdpi.com/1424-8220/22/11/4008>
- 16)<https://link.springer.com/article/10.1007/s11063-023-11204-5>
- 17)<https://www.mdpi.com/1424-8220/22/3/867>
- 18)<https://ieeexplore.ieee.org/abstract/document/9785790>
- 19)<https://www.sciencedirect.com/science/article/abs/pii/S0010482523005462>
- 20)<https://www.nature.com/articles/s41598-022-22644-9>
- 21)<https://www.mdpi.com/1424-8220/22/3/1134>

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22) <https://www.sciencedirect.com/science/article/pii/S1018364721002111>

23) <https://ieeexplore.ieee.org/abstract/document/9113301>

24) <https://link.springer.com/article/10.1186/s12880-020-00534-8>

25) <https://www.mdpi.com/2227-9032/10/12/2481>

26) https://link.springer.com/chapter/10.1007/978-981-15-6067-5_65

27) <https://www.sciencedirect.com/science/article/pii/S2666827021000177>

28) <https://ieeexplore.ieee.org/abstract/document/9397987>

29) <https://link.springer.com/article/10.1007/s11042-018-5714-1>

30) <https://ieeexplore.ieee.org/abstract/document/10776508>